

3. Implement a program for stack that performs following operations using array.

(a) PUSH

(b) POP

(c) PEEP

(d) CHANGE

(e) DISPLAY

```
#include <stdio.h>
```

```
#define STACK_SIZE 5
```

```
int stack[STACK_SIZE];
```

```
int top = -1;
```

```
/* PUSH operation */
```

```
void push()
```

```
{
```

```
    int item;
```

```
    if (top == STACK_SIZE - 1)
```

```
    {
```

```
        printf("Stack Overflow\n");
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Enter element to push: ");
```

```
        scanf("%d", &item);
```

```
        top++;
```

```
        stack[top] = item;
```

```
        printf("Element pushed successfully\n");
```

```
    }
```

```
}
```

```
/* POP operation */
```

```
void pop()
```

```
{
```

```
    if (top == -1)
```

```
    {
```

```
        printf("Stack Underflow\n");
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Popped element: %d\n", stack[top]);
```

```
        top--;
```

```
    }
```

```
}
```

```

/* PEEP operation */
void peep()
{
    int pos;
    if (top == -1)
    {
        printf("Stack is empty\n");
    }
    else
    {
        printf("Enter position from top: ");
        scanf("%d", &pos);

        if (pos <= 0 || pos > top + 1)
        {
            printf("Invalid position\n");
        }
        else
        {
            printf("Element at position %d is %d\n", pos, stack[top - pos + 1]);
        }
    }
}

```

```

/* CHANGE operation */
void change()
{
    int pos, newItem;
    if (top == -1)
    {
        printf("Stack is empty\n");
    }
    else
    {
        printf("Enter position from top: ");
        scanf("%d", &pos);

        if (pos <= 0 || pos > top + 1)
        {
            printf("Invalid position\n");
        }
        else
        {
            printf("Enter new value: ");
            scanf("%d", &newItem);
            stack[top - pos + 1] = newItem;
            printf("Element changed successfully\n");
        }
    }
}

```

```

    }
}

/* DISPLAY operation */
void display()
{
    int i;
    if (top == -1)
    {
        printf("Stack is empty\n");
    }
    else
    {
        printf("Stack elements:\n");
        for (i = top; i >= 0; i--)
        {
            printf("%d\n", stack[i]);
        }
    }
}

/* Main function */
int main()
{
    int choice;
    do
    {
        printf("\n--- STACK OPERATIONS ---\n");
        printf("1. PUSH\n");
        printf("2. POP\n");
        printf("3. PEEP\n");
        printf("4. CHANGE\n");
        printf("5. DISPLAY\n");
        printf("6. EXIT\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1: push(); break;
            case 2: pop(); break;
            case 3: peep(); break;
            case 4: change(); break;
            case 5: display(); break;
            case 6: printf("Exiting program\n"); break;
            default: printf("Invalid choice\n");
        }
    } while (choice != 6);
}

```

```
    return 0;  
}
```